Date:3/2/2023 8:42 AM

810 Vibration Tester Diagnostic Report

Device Serial Number : 3819005
Machine Setup Name : MAY NEN
Measurement Date/Time : 02/28/2023 10:37:34

Drive Train



Maximum Peak : 8.32 mm/sec at 6.38X on 4T in Low Range
1X RPM : 2993 RPM
Overall Vibration : 1.85 g (RMS) @ 3R

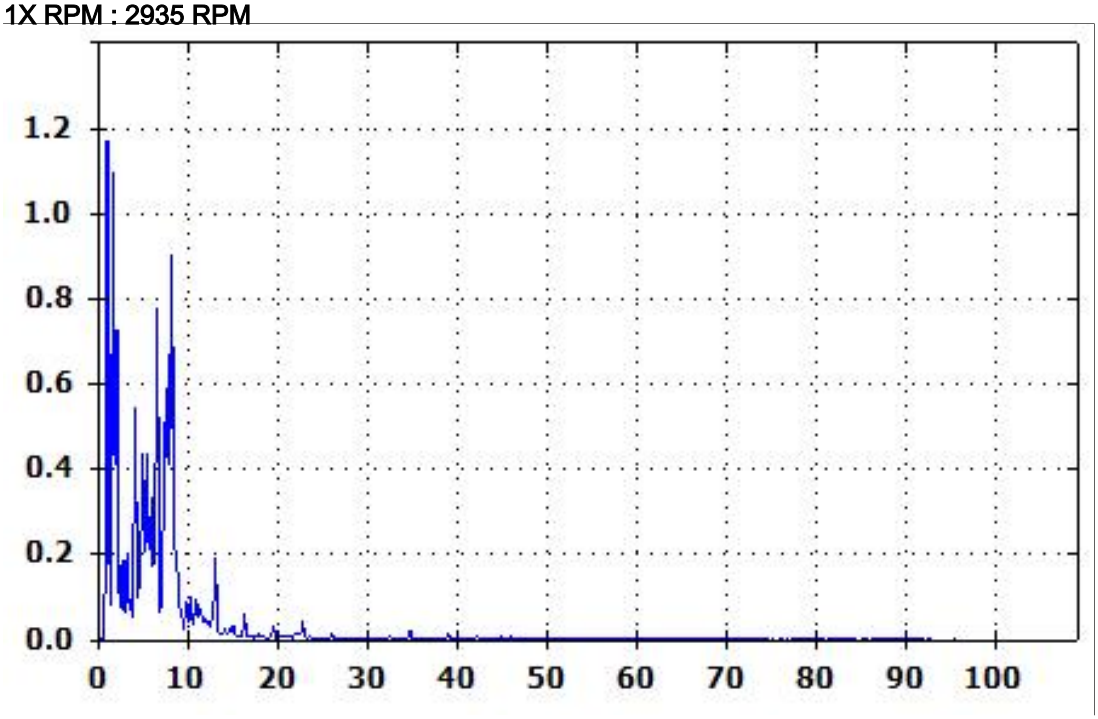
Diagnosis

Fault description	Fault severity	Severity Score	Severity Scale
Pump Free End Bearing Wear	Extreme	77/100	
Pump Drive End Bearing Wear	Serious	59/100	
Motor Drive End Bearing Wear	Serious	59/100	
Pump Rotor Or Bearing Looseness	Serious	52/100	
Motor Drive End Bearing Looseness	Moderate	45/100	
Indication Of Possible Coupling Wear Or Looseness	Moderate	42/100	
Motor Free End Bearing Wear	Moderate	32/100	
Motor Free End Bearing Looseness	Moderate	32/100	

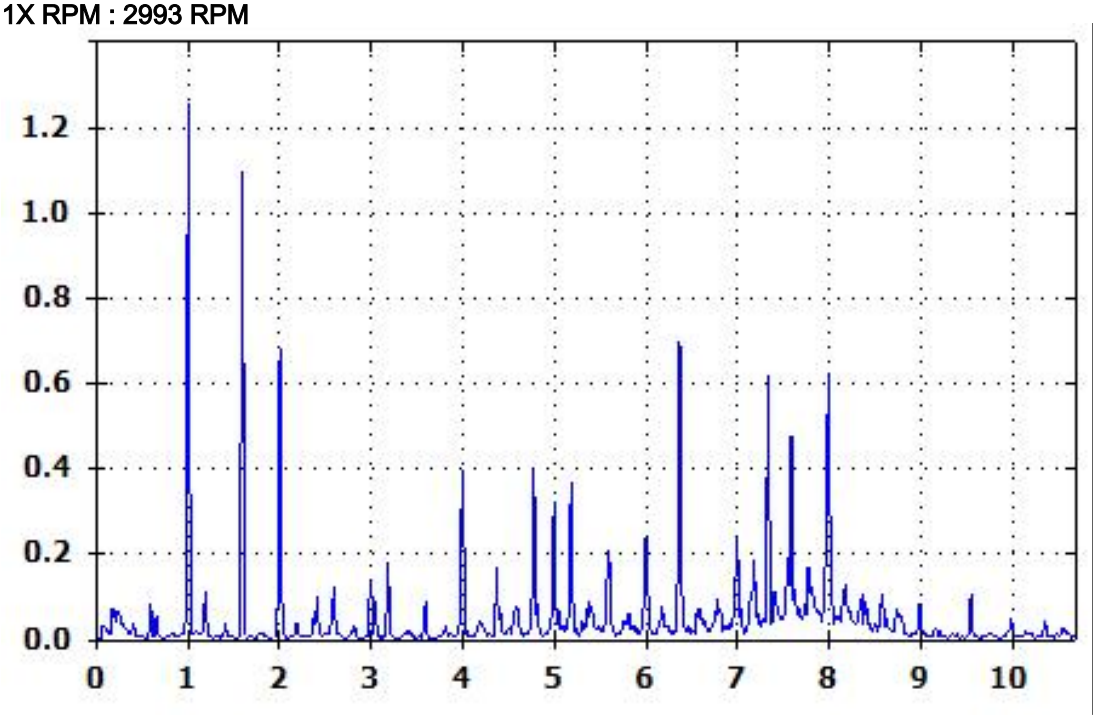
Recommendations

Recommendations	Priority	Priority Description
Specialized analysis or consulting recommended due to a non-standard fault	4	Mandatory
Monitor for increased vibration	2	Desirable
Replace Motor Bearings	2	Desirable

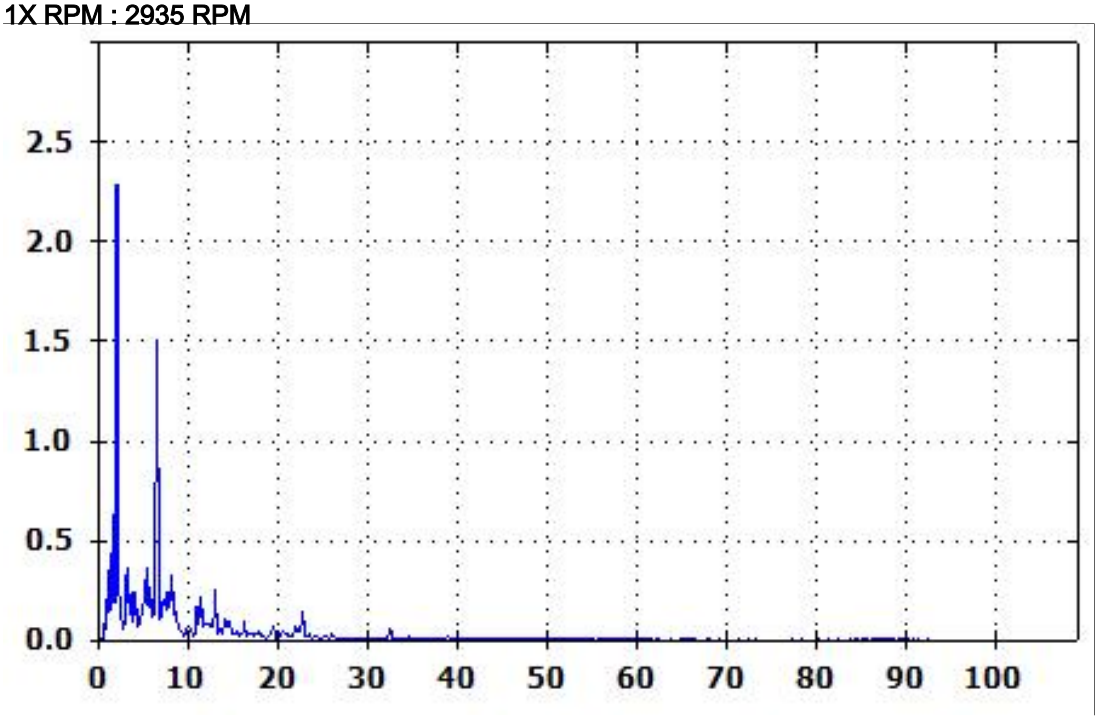
Location : Location 1
Range : High range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2935 RPM



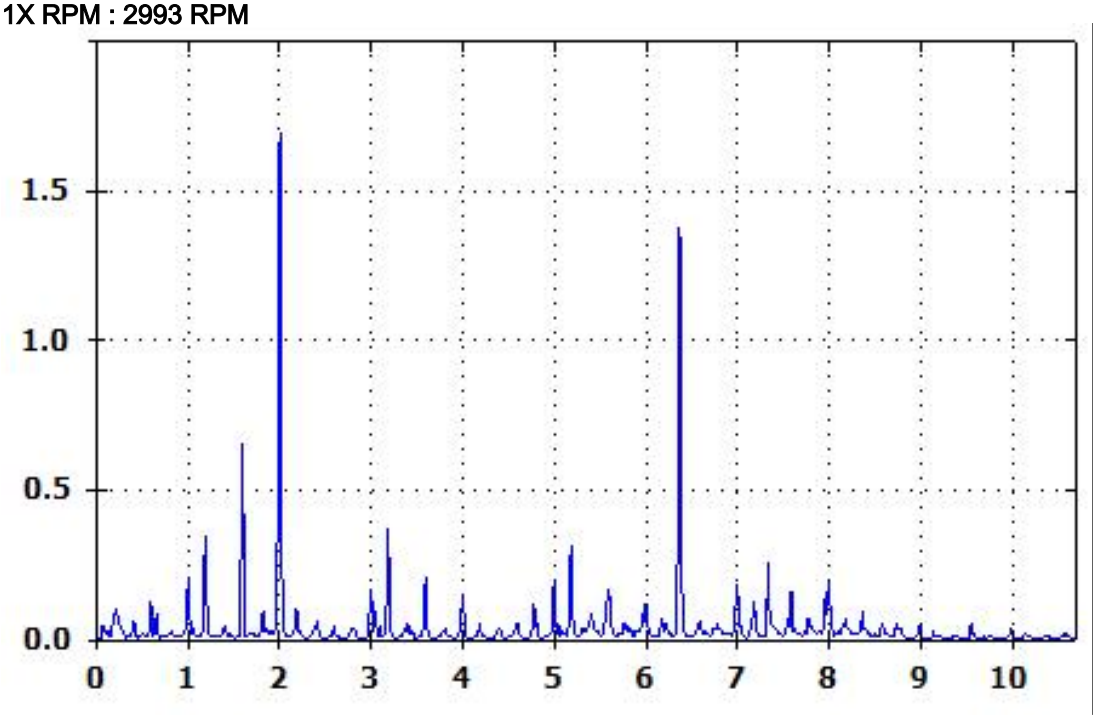
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Range : Low range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2993 RPM



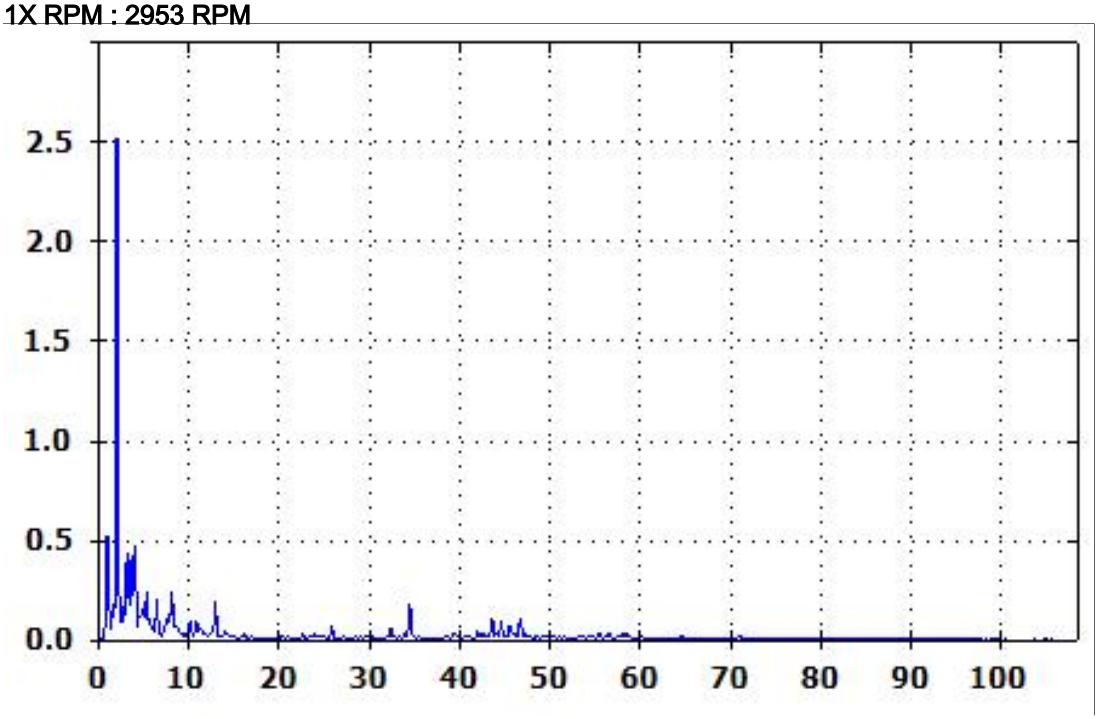
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Range : High range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2935 RPM



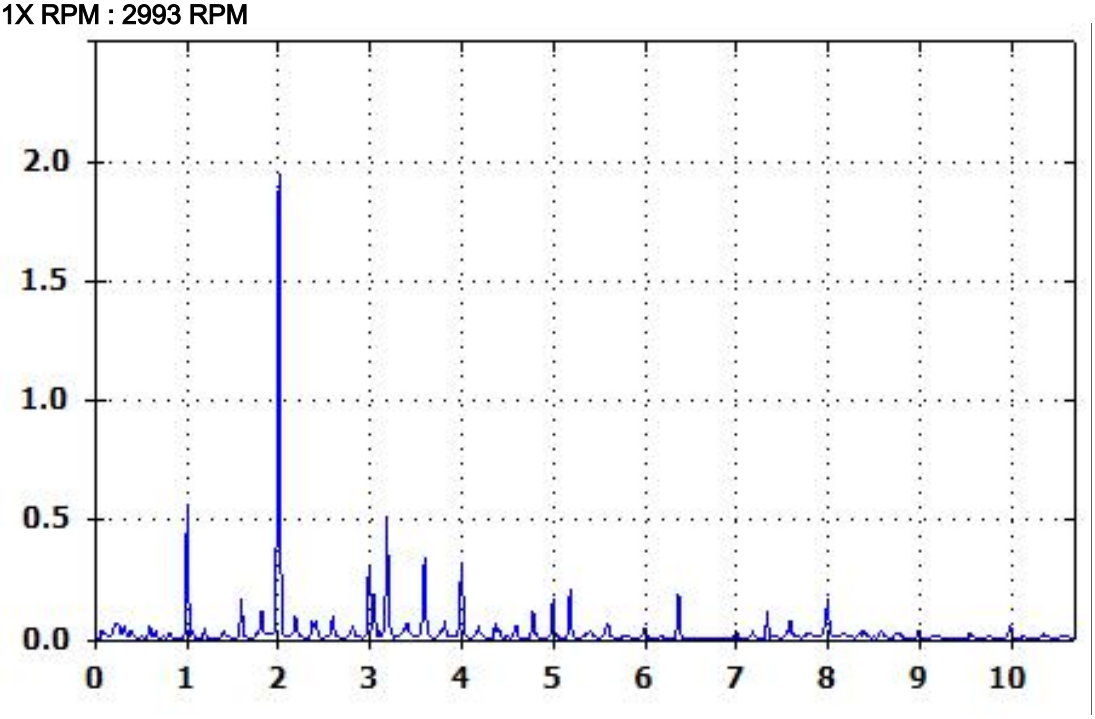
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Range : Low range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2993 RPM



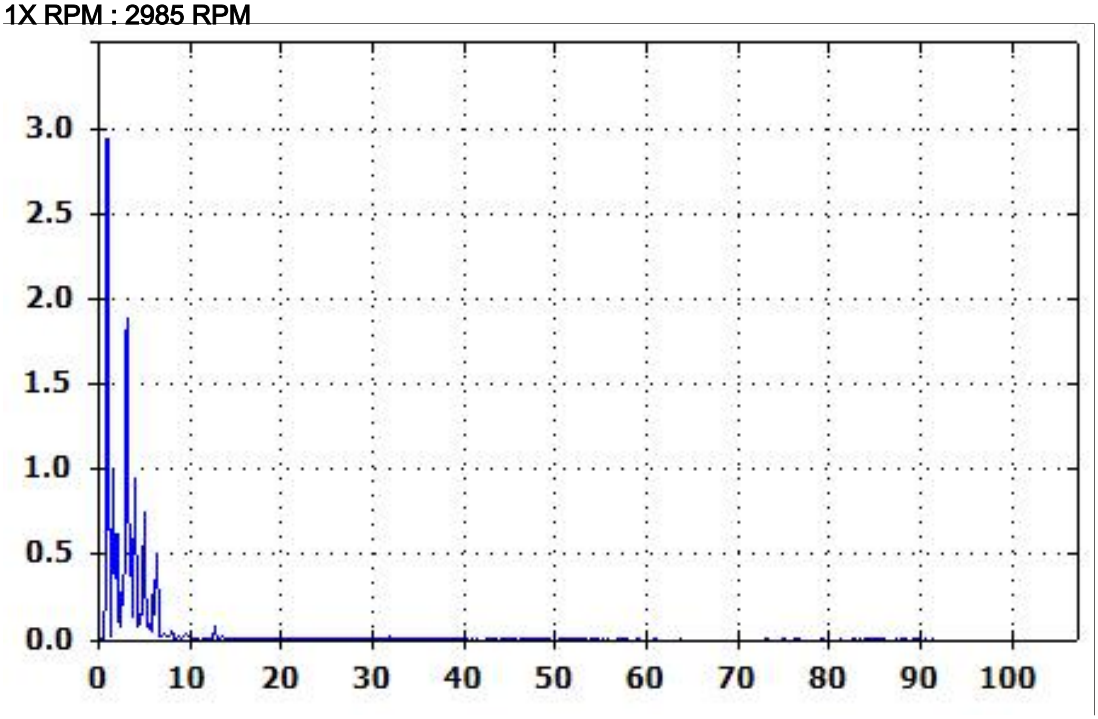
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Range : High range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2953 RPM



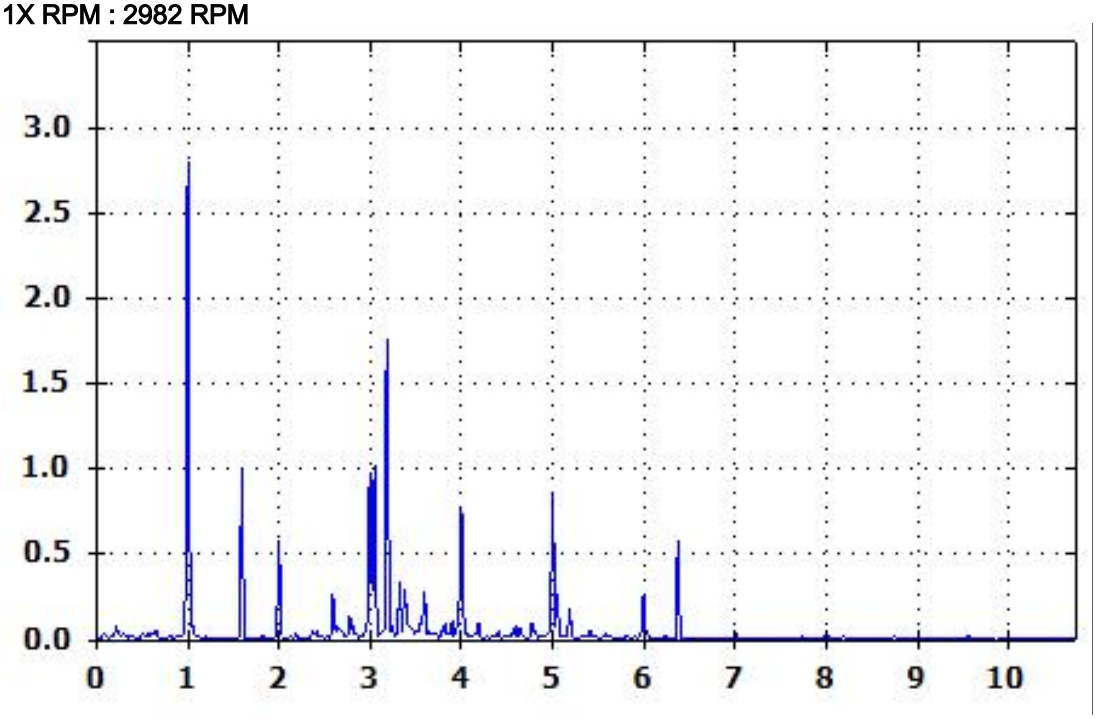
Location : Location 1
Range : Low range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2993 RPM



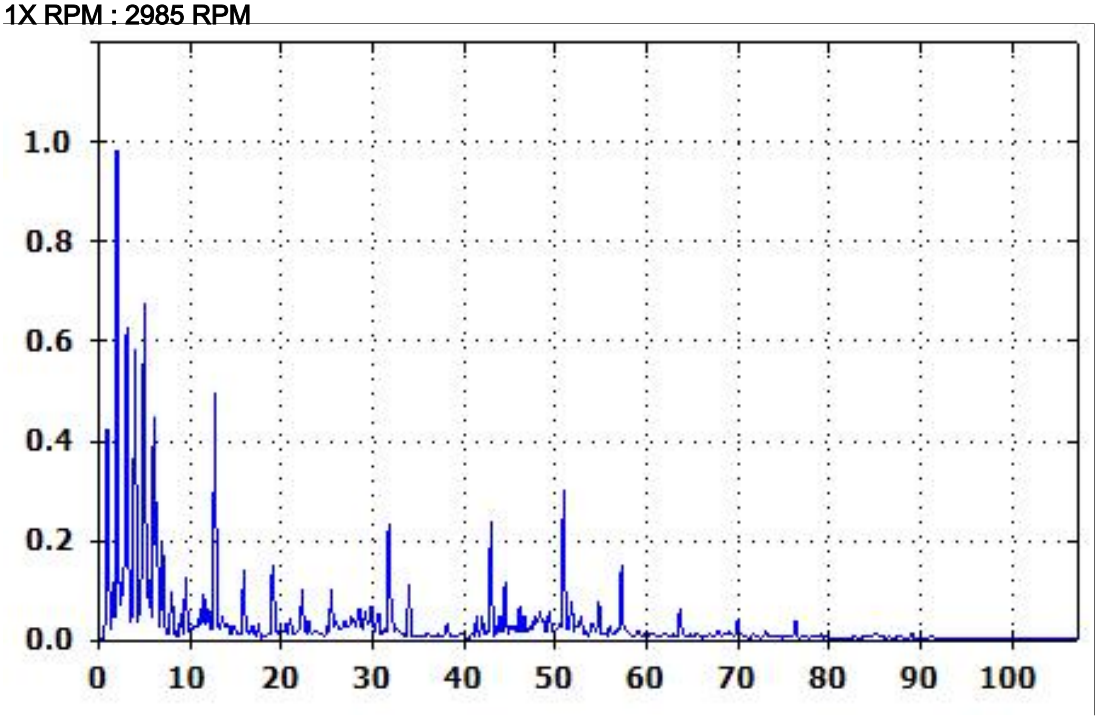
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Range : High range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2985 RPM



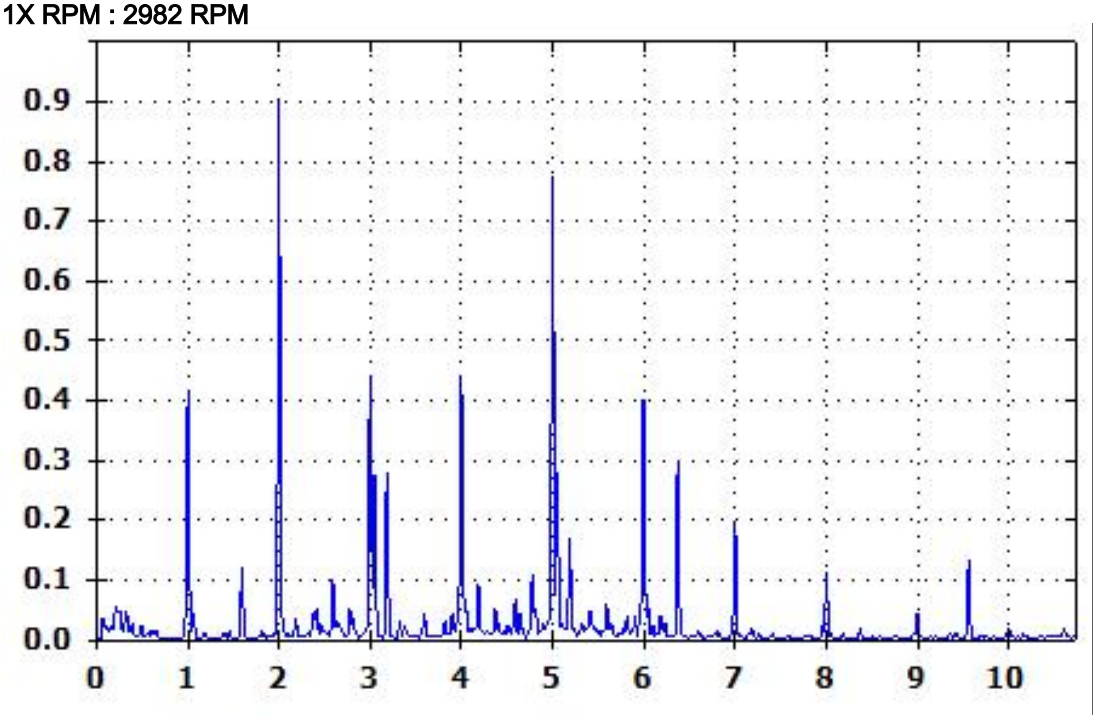
Location : Location 2
Range : Low range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2982 RPM



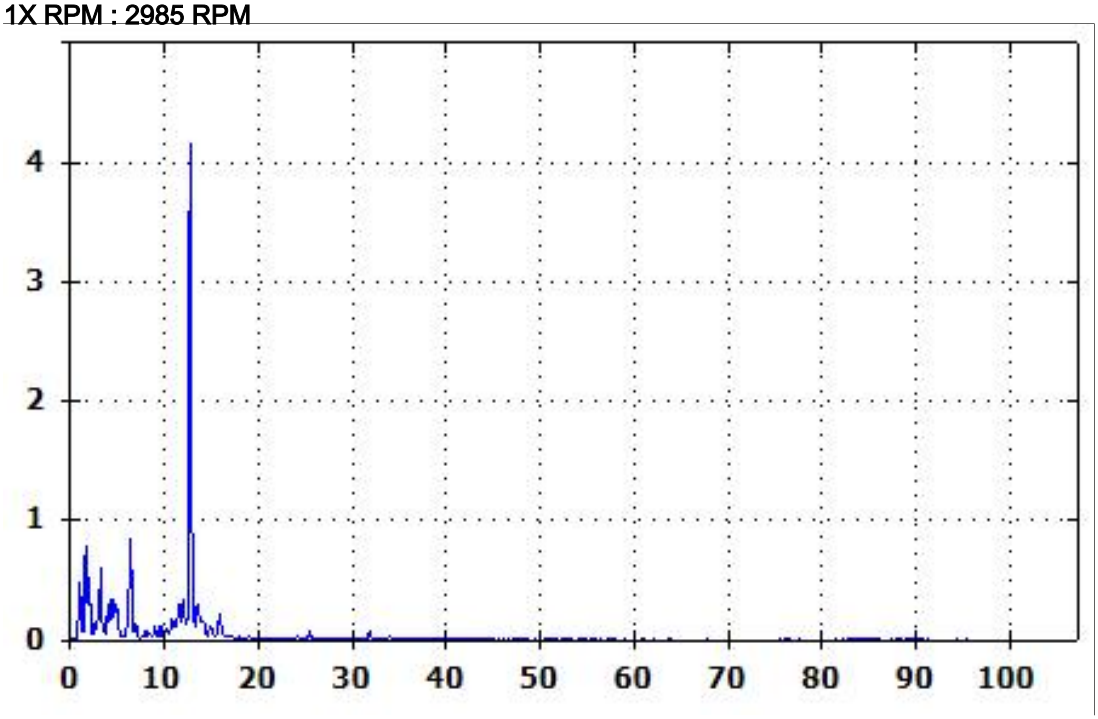
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Range : High range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2985 RPM



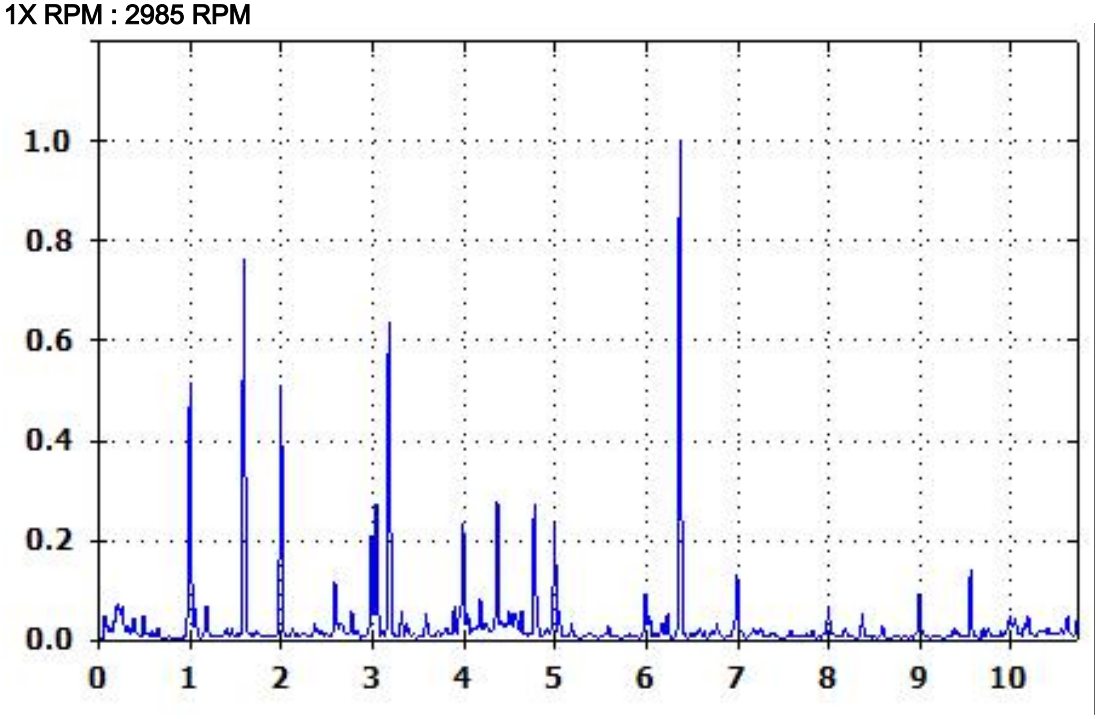
Location : Location 2
Range : Low range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2982 RPM



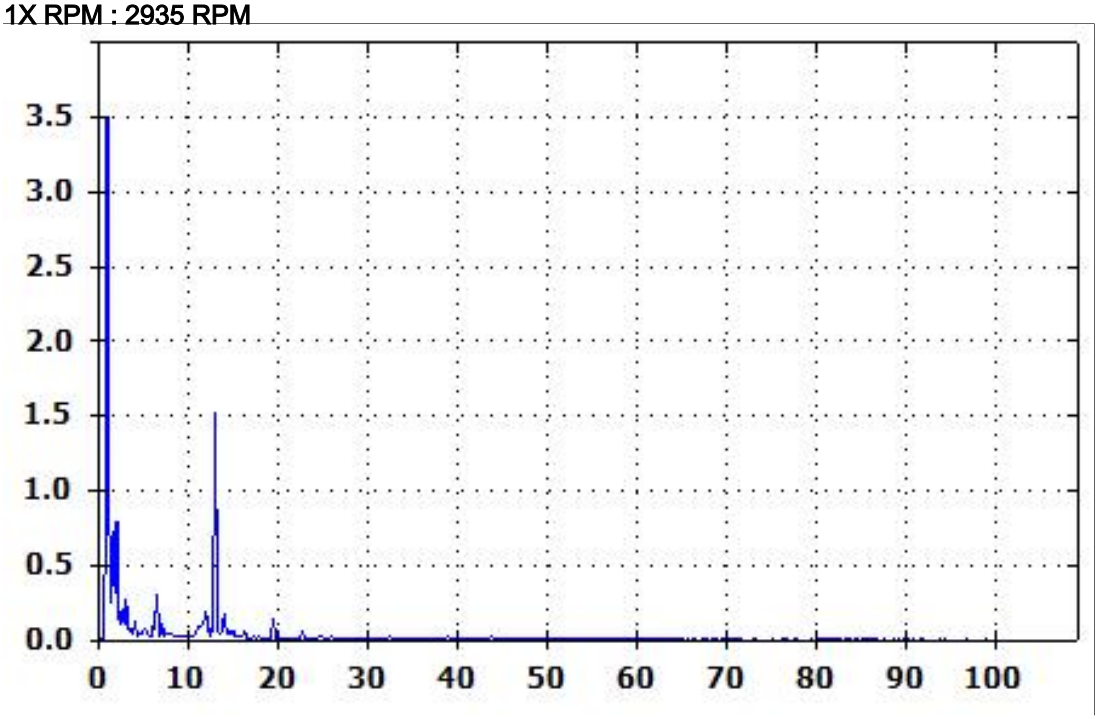
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Range : High range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2985 RPM



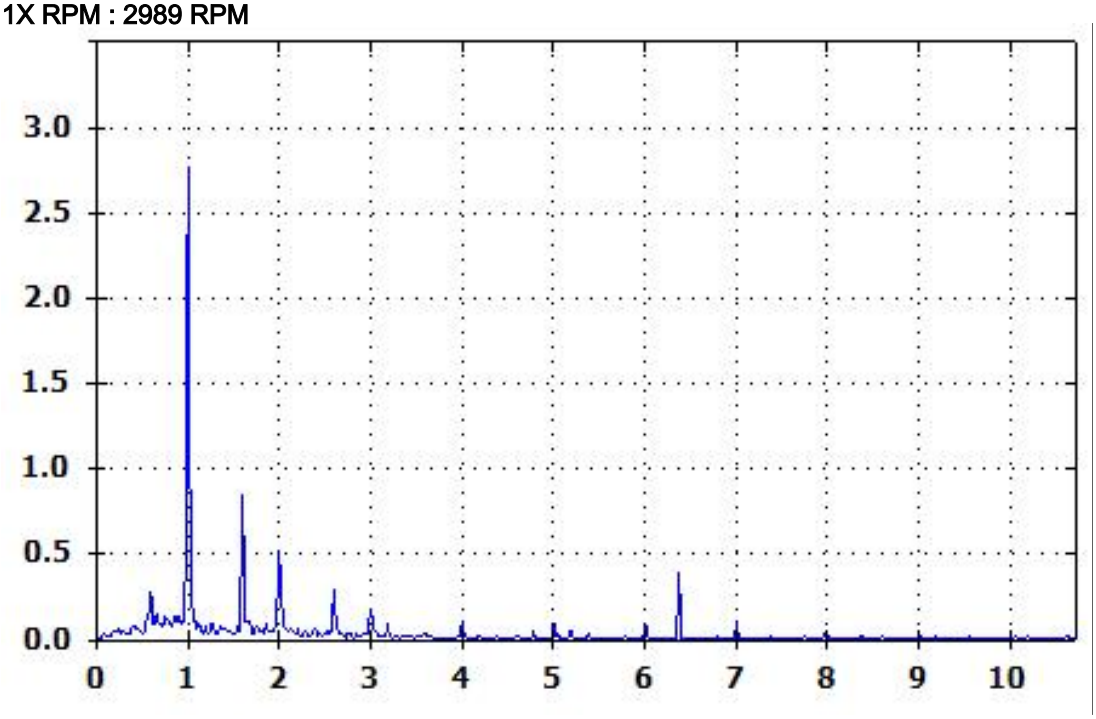
Location : Location 2
Range : Low range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2985 RPM



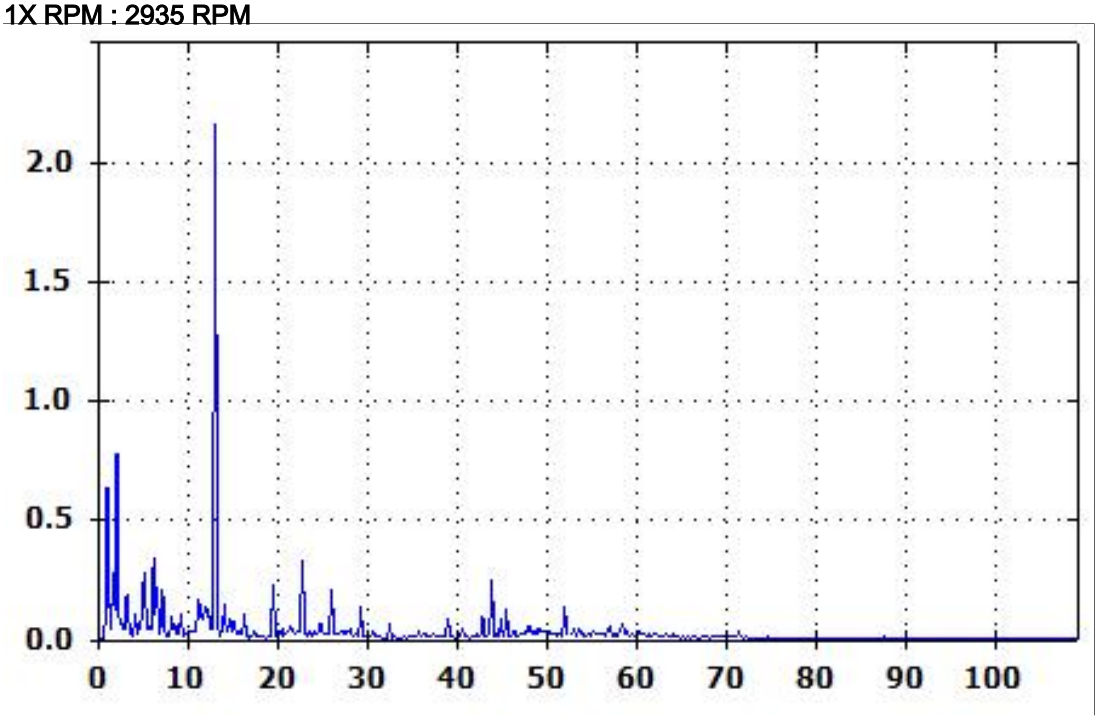
Location : Location 3
Range : High range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2935 RPM



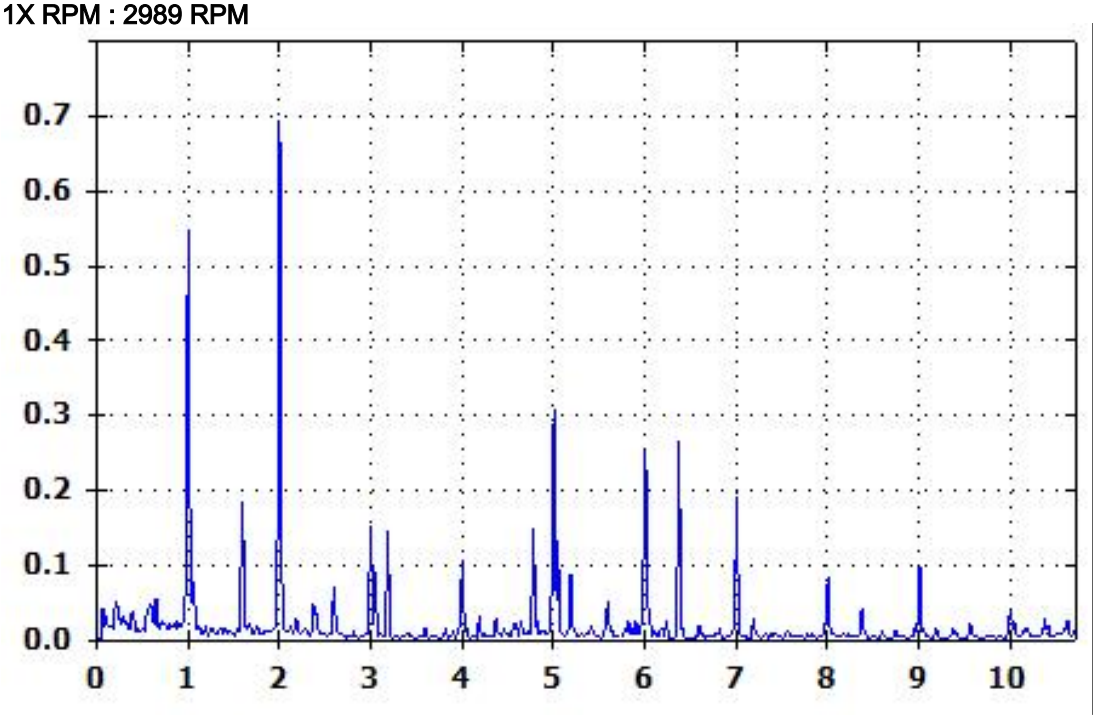
Location : Location 3
Range : Low range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2989 RPM



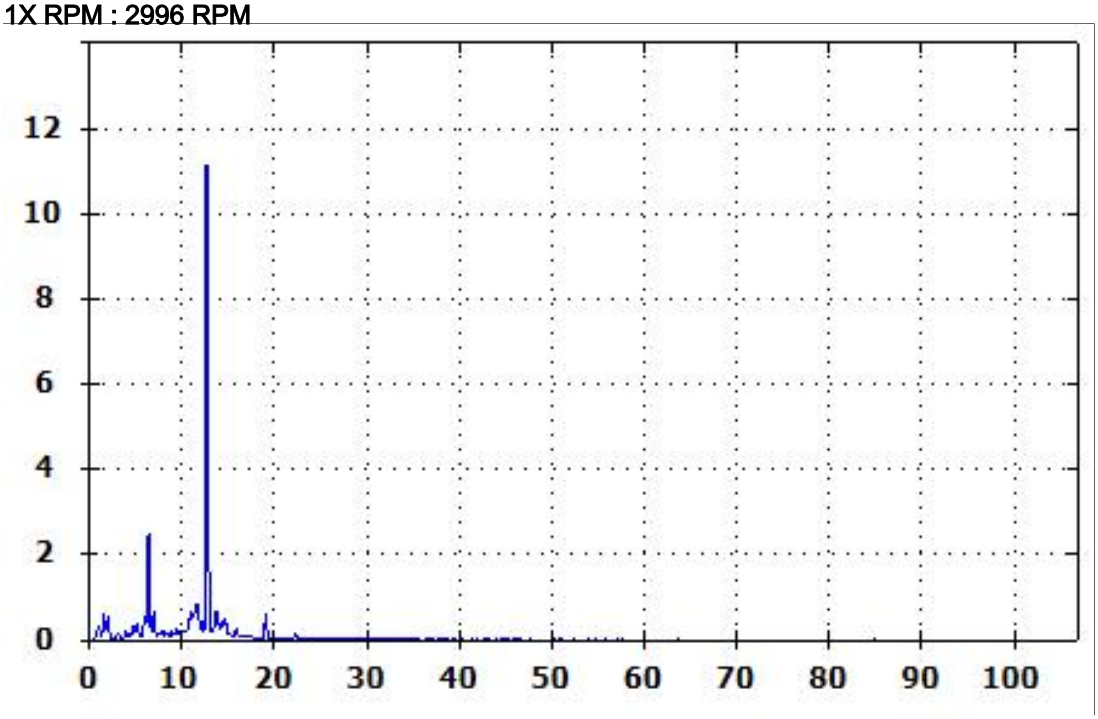
Location : Location 3
Range : High range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2935 RPM



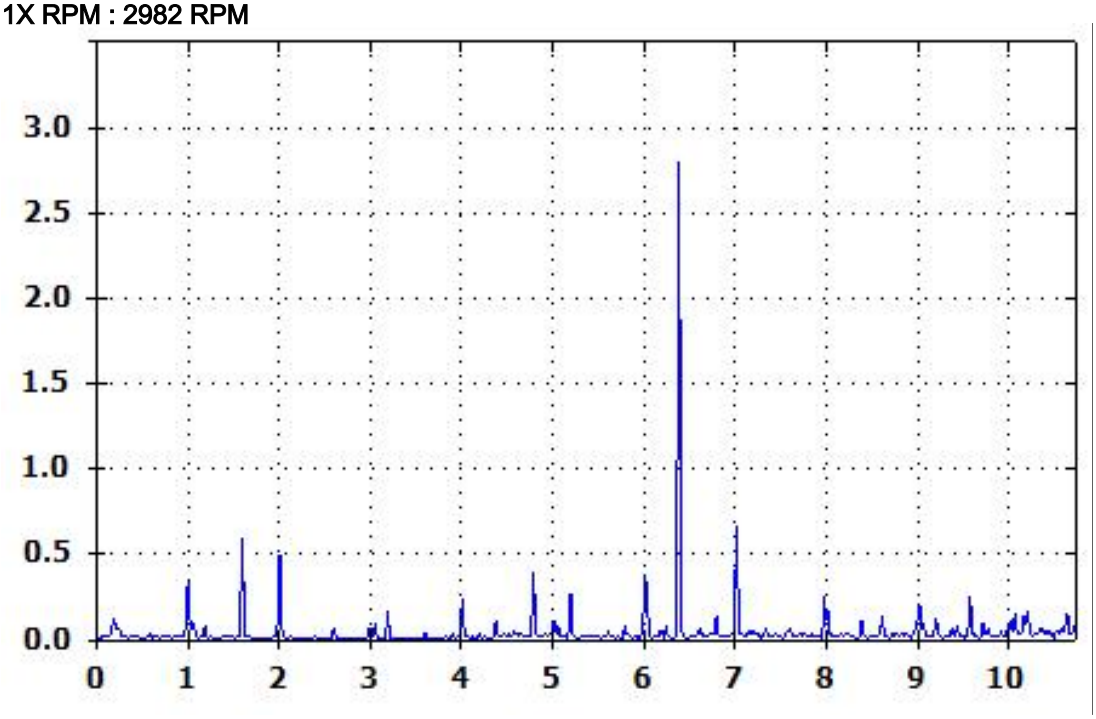
Location : Location 3
Range : Low range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2989 RPM



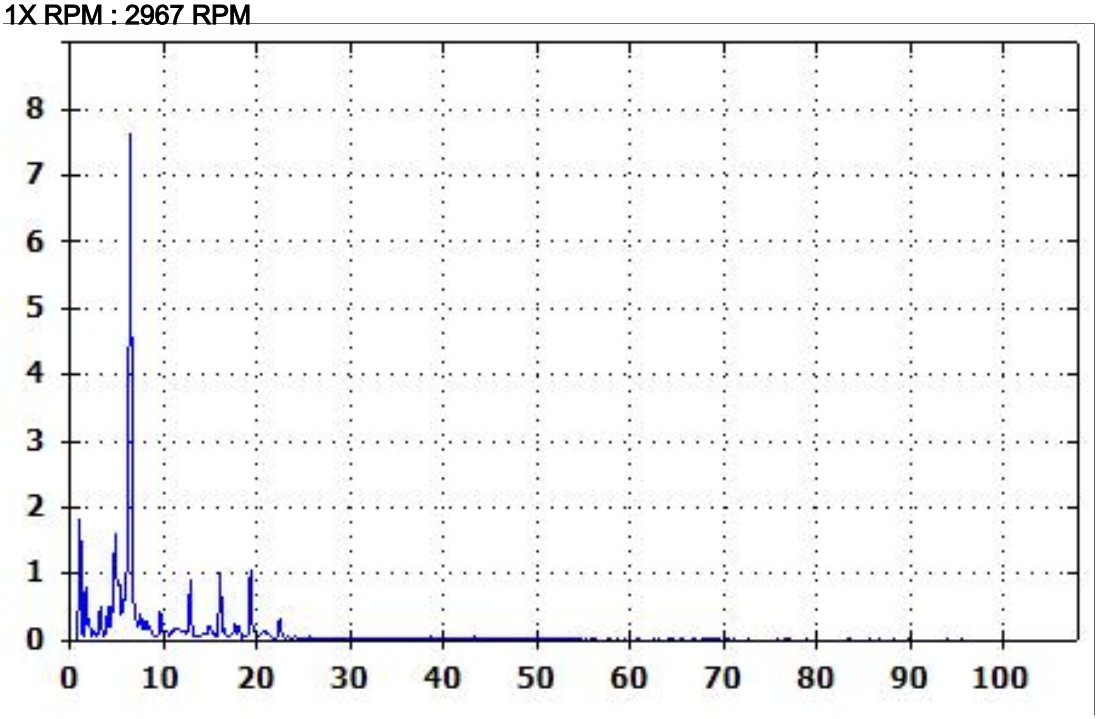
Location : Location 3
Range : High range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2996 RPM



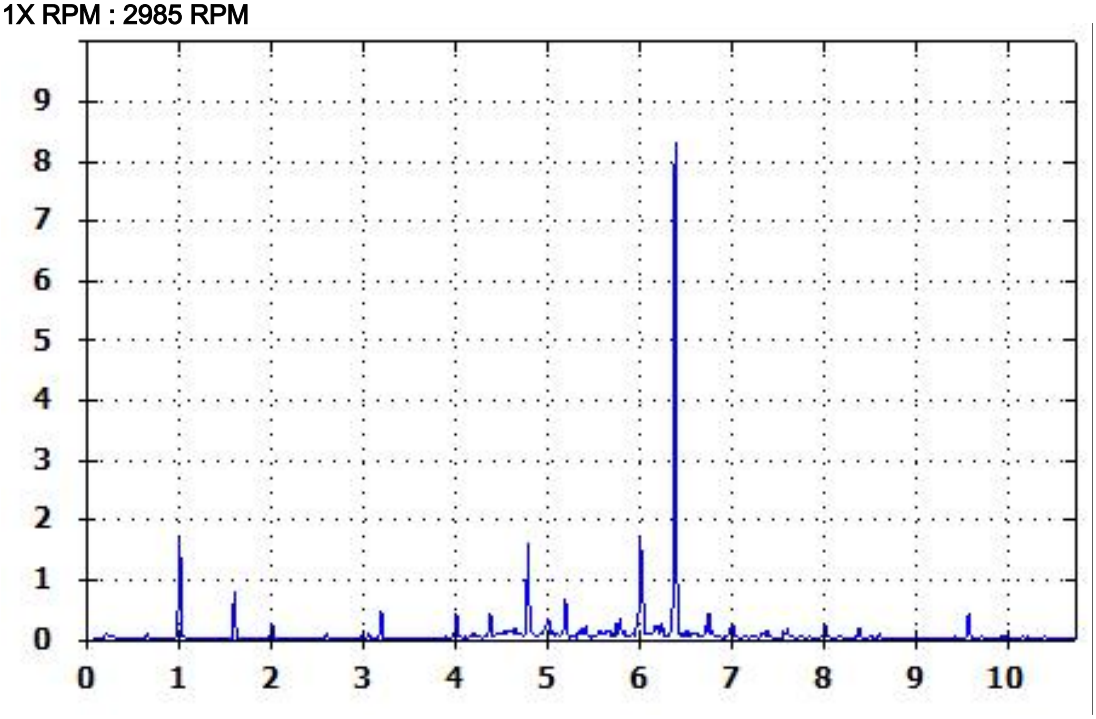
Location : Location 3
Range : Low range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2982 RPM



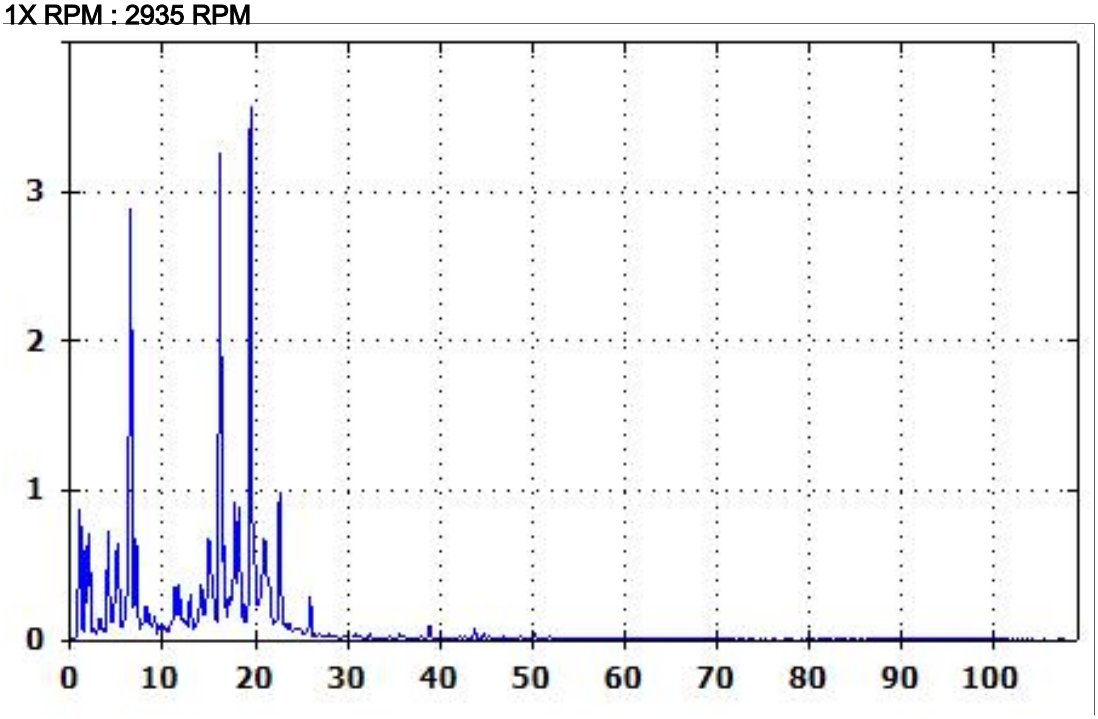
Location : Location 4
Range : High range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2967 RPM



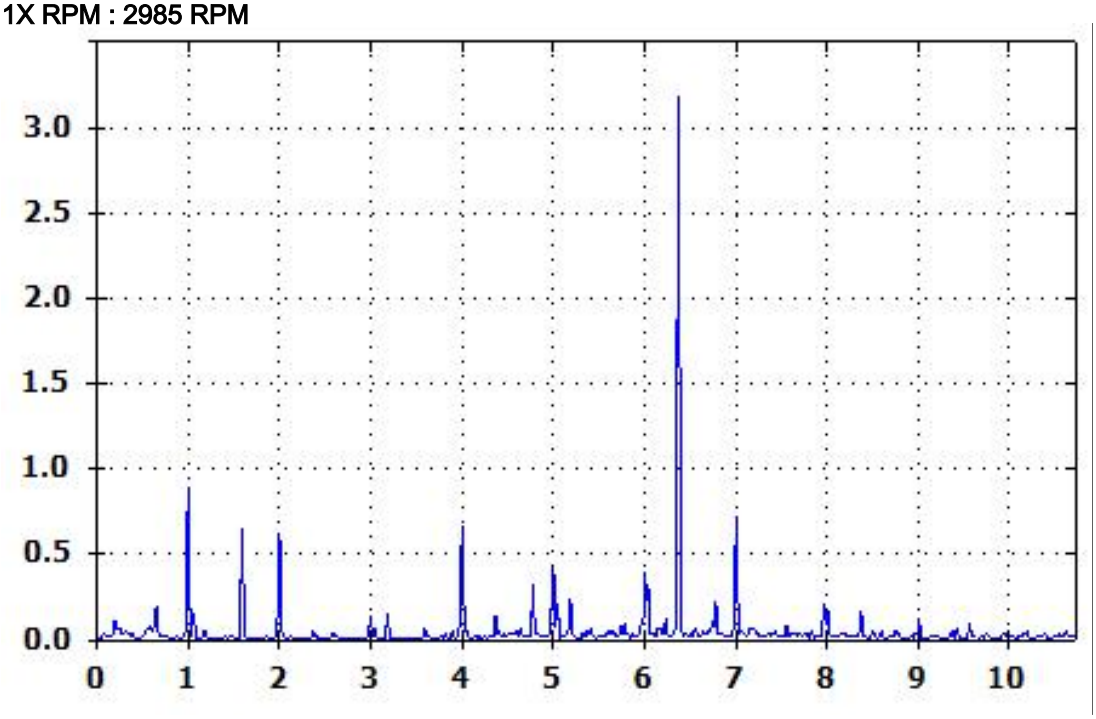
Location : Location 4
Range : Low range
Axis : Axial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2985 RPM



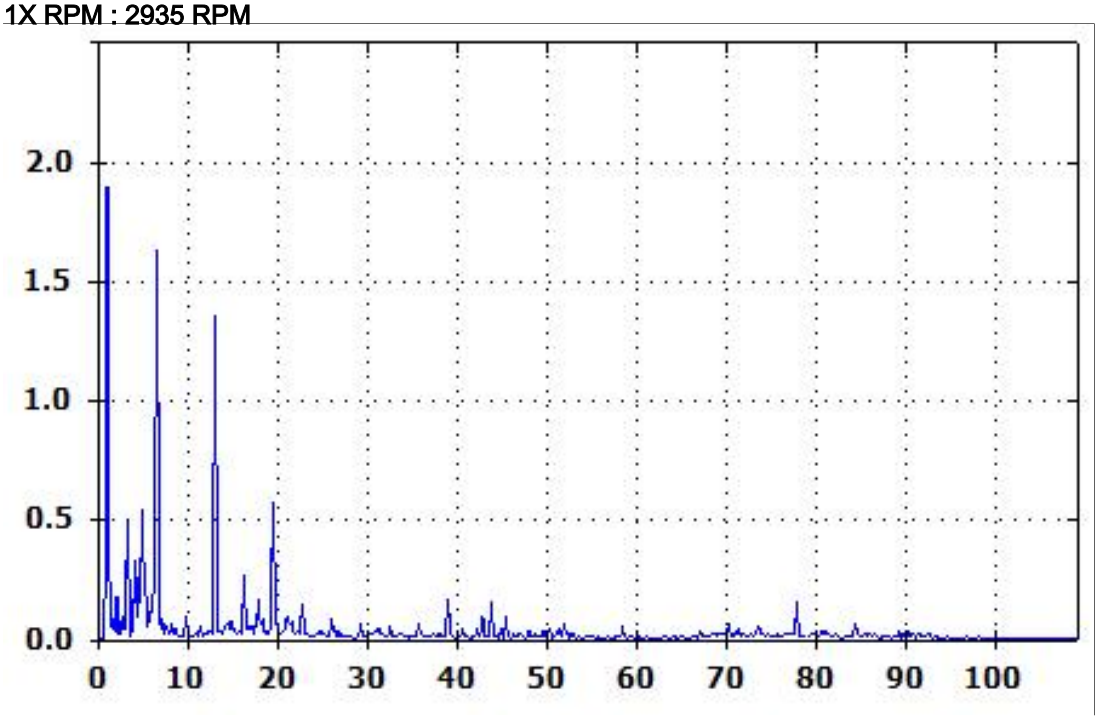
Location : Location 4
Range : High range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2935 RPM



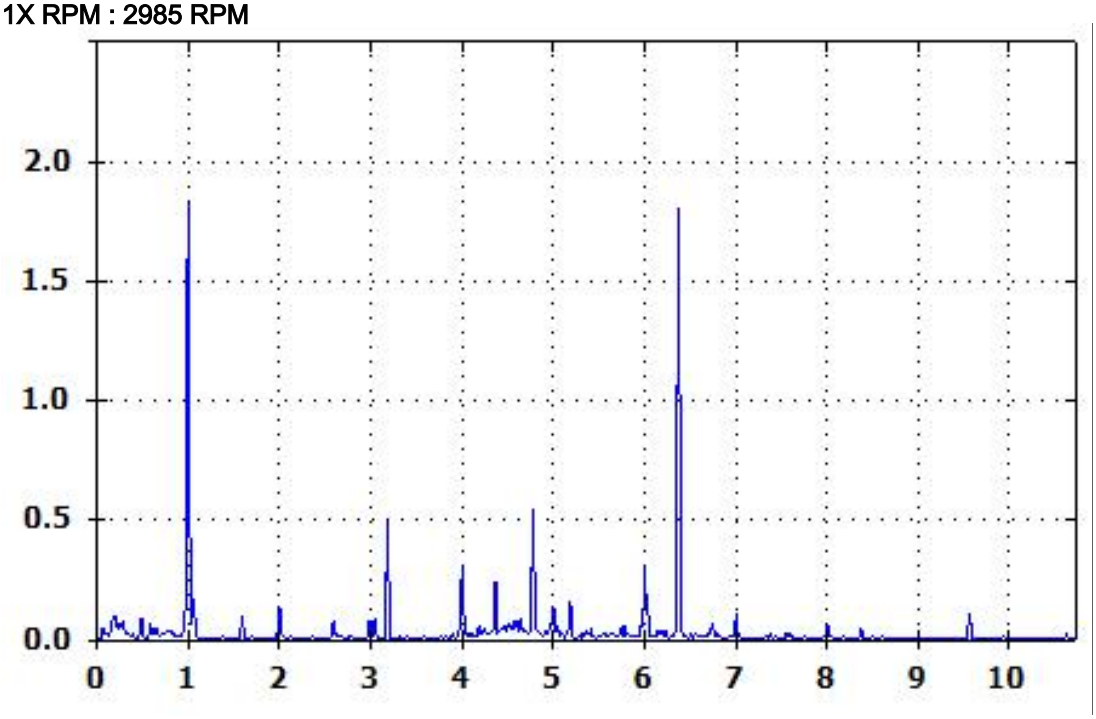
Location : Location 4
Range : Low range
Axis : Tangential
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2985 RPM



Location : Location 4
Range : High range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2935 RPM



Location : Location 4
Range : Low range
Axis : Radial
X Axis Unit : Orders
Y Axis Unit : mm/sec
1X RPM : 2985 RPM



Fault description : Pump Free End Bearing Wear

Fault severity : Extreme (77)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location4	Axial	8.32 mm/sec	6.37	Low
Cited peak2	Location4	Tangential	3.58 mm/sec	19.5	High
Cited peak3	Location4	Tangential	3.26 mm/sec	16.2	High
Cited peak4	Location4	Tangential	3.17 mm/sec	6.37	Low
Cited peak5	Location4	Radial	1.81 mm/sec	6.37	Low
Cited peak6	Location4	Axial	1.59 mm/sec	4.78	Low
Cited peak7	Location4	Axial	1.05 mm/sec	19.2	High
Cited peak8	Location4	Axial	1.01 mm/sec	16	High
Cited peak9	Location4	Radial	0.58 mm/sec	19.5	High
Cited peak10	Location4	Radial	0.55 mm/sec	4.78	Low

Fault description : Pump Drive End Bearing Wear

Fault severity : Serious (59)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location3	Radial	11.14 mm/sec	12.7	High
Cited peak2	Location3	Radial	2.79 mm/sec	6.39	Low
Cited peak3	Location3	Radial	0.87 mm/sec	11.7	High
Cited peak4	Location3	Axial	0.39 mm/sec	6.37	Low
Cited peak5	Location3	Tangential	0.34 mm/sec	22.7	High
Cited peak6	Location3	Tangential	0.27 mm/sec	6.37	Low

Fault description : Motor Drive End Bearing Wear

Fault severity : Serious (59)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location2	Radial	4.16 mm/sec	12.7	High
Cited peak2	Location2	Axial	1.77 mm/sec	3.19	Low
Cited peak3	Location2	Axial	1.02 mm/sec	3.05	Low
Cited peak4	Location2	Radial	1.00 mm/sec	6.37	Low
Cited peak5	Location2	Tangential	0.50 mm/sec	12.7	High
Cited peak6	Location2	Tangential	0.30 mm/sec	51	High
Cited peak7	Location2	Tangential	0.30 mm/sec	6.37	Low
Cited peak8	Location2	Tangential	0.28 mm/sec	3.05	Low
Cited peak9	Location2	Axial	0.08 mm/sec	12.7	High

Fault description : Pump Rotor Or Bearing Looseness

Fault severity : Serious (52)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location3	Tangential	2.16 mm/sec	13	High
Cited peak2	Location4	Axial	1.72 mm/sec	6	Low
Cited peak3	Location3	Axial	1.52 mm/sec	13	High
Cited peak4	Location4	Radial	1.36 mm/sec	13	High
Cited peak5	Location4	Axial	1.01 mm/sec	16	High
Cited peak6	Location4	Tangential	0.72 mm/sec	7	Low
Cited peak7	Location4	Tangential	0.67 mm/sec	4	Low
Cited peak8	Location3	Radial	0.67 mm/sec	7	Low
Cited peak9	Location4	Axial	0.44 mm/sec	4	Low
Cited peak10	Location4	Tangential	0.43 mm/sec	5	Low
Cited peak11	Location4	Tangential	0.39 mm/sec	6	Low
Cited peak12	Location3	Radial	0.38 mm/sec	6	Low
Cited peak13	Location4	Axial	0.36 mm/sec	5	Low
Cited peak14	Location4	Radial	0.31 mm/sec	4	Low
Cited peak15	Location4	Radial	0.31 mm/sec	6	Low
Cited peak16	Location3	Tangential	0.31 mm/sec	5	Low
Cited peak17	Location3	Tangential	0.25 mm/sec	6	Low
Cited peak18	Location3	Radial	0.25 mm/sec	8	Low
Cited peak19	Location4	Axial	0.25 mm/sec	8	Low
Cited peak20	Location4	Axial	0.24 mm/sec	7	Low
Cited peak21	Location3	Radial	0.23 mm/sec	4	Low
Cited peak22	Location4	Tangential	0.22 mm/sec	8	Low
Cited peak23	Location3	Radial	0.22 mm/sec	9	Low
Cited peak24	Location3	Axial	0.19 mm/sec	12	High
Cited peak25	Location3	Tangential	0.19 mm/sec	7	Low
Cited peak26	Location3	Axial	0.18 mm/sec	3	Low
Cited peak27	Location4	Tangential	0.12 mm/sec	9	Low
Cited peak28	Location4	Radial	0.11 mm/sec	7	Low
Cited peak29	Location3	Axial	0.11 mm/sec	7	Low

Fault description : Motor Drive End Bearing Looseness

Fault severity : Moderate (45)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location2	Axial	1.02 mm/sec	3	Low
Cited peak2	Location2	Axial	0.87 mm/sec	5	Low
Cited peak3	Location2	Axial	0.79 mm/sec	4	Low
Cited peak4	Location2	Tangential	0.78 mm/sec	5	Low
Cited peak5	Location2	Tangential	0.44 mm/sec	4	Low
Cited peak6	Location2	Tangential	0.44 mm/sec	3	Low
Cited peak7	Location2	Tangential	0.40 mm/sec	6	Low
Cited peak8	Location2	Radial	0.34 mm/sec	12	High
Cited peak9	Location2	Radial	0.27 mm/sec	3	Low
Cited peak10	Location2	Axial	0.26 mm/sec	6	Low
Cited peak11	Location2	Radial	0.24 mm/sec	4	Low
Cited peak12	Location2	Radial	0.23 mm/sec	5	Low
Cited peak13	Location2	Tangential	0.20 mm/sec	7	Low
Cited peak14	Location2	Radial	0.13 mm/sec	7	Low
Cited peak15	Location2	Tangential	0.11 mm/sec	8	Low

Fault description : Indication Of Possible Coupling Wear Or Looseness

Fault severity : Moderate (42)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location3	Tangential	2.16 mm/sec	13	High
Cited peak2	Location3	Axial	1.52 mm/sec	13	High
Cited peak3	Location2	Axial	1.02 mm/sec	3	Low
Cited peak4	Location2	Axial	0.87 mm/sec	5	Low
Cited peak5	Location2	Axial	0.79 mm/sec	4	Low
Cited peak6	Location2	Tangential	0.78 mm/sec	5	Low
Cited peak7	Location3	Radial	0.67 mm/sec	7	Low
Cited peak8	Location2	Tangential	0.44 mm/sec	4	Low
Cited peak9	Location2	Tangential	0.44 mm/sec	3	Low
Cited peak10	Location2	Tangential	0.40 mm/sec	6	Low
Cited peak11	Location3	Radial	0.38 mm/sec	6	Low
Cited peak12	Location3	Tangential	0.31 mm/sec	5	Low
Cited peak13	Location2	Radial	0.27 mm/sec	3	Low
Cited peak14	Location2	Axial	0.26 mm/sec	6	Low
Cited peak15	Location3	Tangential	0.25 mm/sec	6	Low
Cited peak16	Location3	Radial	0.25 mm/sec	8	Low
Cited peak17	Location2	Radial	0.24 mm/sec	4	Low
Cited peak18	Location2	Radial	0.23 mm/sec	5	Low
Cited peak19	Location3	Radial	0.23 mm/sec	4	Low
Cited peak20	Location2	Tangential	0.20 mm/sec	7	Low
Cited peak21	Location3	Tangential	0.19 mm/sec	7	Low
Cited peak22	Location3	Axial	0.18 mm/sec	3	Low
Cited peak23	Location2	Radial	0.13 mm/sec	7	Low
Cited peak24	Location2	Tangential	0.11 mm/sec	8	Low
Cited peak25	Location3	Axial	0.11 mm/sec	7	Low

Fault description : Motor Free End Bearing Wear

Fault severity : Moderate (32)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location1	Tangential	1.38 mm/sec	6.37	Low
Cited peak2	Location1	Axial	0.70 mm/sec	6.37	Low
Cited peak3	Location1	Radial	0.51 mm/sec	3.18	Low
Cited peak4	Location1	Tangential	0.21 mm/sec	11.4	High
Cited peak5	Location1	Radial	0.18 mm/sec	34.5	High
Cited peak6	Location1	Axial	0.07 mm/sec	11.4	High

Fault description : Motor Free End Bearing Looseness

Fault severity : Moderate (32)

Cited peak number	Bearing	Axis	Vibration amplitude	Order	Range
Cited peak1	Location1	Axial	0.62 mm/sec	8	Low
Cited peak2	Location1	Axial	0.40 mm/sec	4	Low
Cited peak3	Location1	Axial	0.33 mm/sec	5	Low
Cited peak4	Location1	Radial	0.32 mm/sec	4	Low
Cited peak5	Location1	Radial	0.30 mm/sec	3	Low
Cited peak6	Location1	Tangential	0.25 mm/sec	13	High
Cited peak7	Location1	Axial	0.24 mm/sec	7	Low
Cited peak8	Location1	Axial	0.24 mm/sec	6	Low
Cited peak9	Location1	Tangential	0.21 mm/sec	5	Low
Cited peak10	Location1	Tangential	0.19 mm/sec	8	Low
Cited peak11	Location1	Axial	0.19 mm/sec	13	High
Cited peak12	Location1	Tangential	0.19 mm/sec	7	Low
Cited peak13	Location1	Radial	0.19 mm/sec	13	High
Cited peak14	Location1	Radial	0.17 mm/sec	8	Low

Machine Setup Details

Machine Setup Name : MAY NEN

Setup Field	Input
Motor type	AC
AC motor with VFD	No
Speed in RPM	2970
Nominal kw	75
Motor mounted	Horizontal
Motor has	Roller bearing
Motor detached from drive train	No
Motor close-coupled	No
Coupling between motor and next component	Yes
Next component	Pump
Driven component bearing type	Roller bearing
Pump type	Gear/screw
No. of teeth/lobes [optional]	